6

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards.

(BL2,BL6)

Assessment Tool Weightage: 30%

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding, some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

DETAILED SOLUTIONS — INDUSTRY PROBLEM-SOLVING TASK

Each of the five industry scenarios below is expanded into a complete network-design solution: a problem analysis, the proposed topology with a labelled diagram, a device/media specification, an addressing or configuration scheme where applicable, the technical justification, and a testing and validation plan — directly addressing every rubric criterion (Problem Analysis, Solution Design, Technical Implementation, Use of Tools & Technologies, Problem-Solving Approach, Innovation, Testing & Validation, and Documentation).

Solution 1 — Star Topology for a 20-Person Startup

1.1 Problem Analysis

Requirements: 20 employees, a mix of workstations, network printers, and VoIP phones, a tight IT budget typical of a startup, need for simple day-to-day management by a small (often part-time) IT staff, and room to grow to roughly 30 devices without a redesign. Downtime tolerance is moderate — brief single-user outages are acceptable, but a network-wide outage would halt the whole office.

1.2 Proposed Topology and Network Diagram

A pure star topology is proposed, with a single Layer 2/Layer 3 managed switch as the central point of connection for every device, and the switch uplinked to a router/firewall for internet access.

```
              [ Internet ]
                   |
        [ Router / Firewall ]
            (pfSense / Cisco RV)
                   |
     [ 24-Port L2/L3 Managed Switch ]
        /       |       |      |       \
    Wi-Fi   Printer  VoIP  Desk-1 ... Desk-19
     AP              Phone
```

1.3 Device and Cable Specification

Item	Model / Type	Qty	Purpose
Core switch	Cisco Catalyst 1300 / Netgear ProSafe, 24-port, L2/L3 managed	1	Central connection point; VLAN & QoS capable
Router / firewall	Cisco RV series or pfSense appliance	1	Internet uplink, DHCP, NAT, firewall rules
Wireless AP	802.11ac/ax access point	1-2	Wi-Fi coverage for laptops & mobiles
Workstations	Desktop / laptop PCs	17-19	End-user devices
Network printer	Networked laser/MFP printer	1-2	Shared printing
VoIP phones	SIP desk phones	up to 20	Voice traffic (QoS-tagged)
Patch cabling	Cat6 UTP, RJ-45	~22	Gigabit link, ≤100 m switch-to-device

1.4 Sample Cable-Run Table (Switch to Each Device)

```
Host          Central Switch     Cat6 Cable Length (m)
-----------------------------------------------------
Desk-1        Switch-Core-A1     8.5
Desk-2        Switch-Core-A1     11.2
Desk-3        Switch-Core-A1     15.0
...           ...                ...
Printer-1     Switch-Core-A1     6.0
VoIP-1        Switch-Core-A1     9.4
AP-1          Switch-Core-A1     22.7
-----------------------------------------------------
All runs <= 100 m -> compliant with IEEE 802.3 / TIA-568
```

1.5 IP Addressing Plan

VLAN / Pool	Subnet	Usable Hosts	Assigned To
Data VLAN 10	192.168.10.0/24	254	Workstations & printers
Voice VLAN 20	192.168.20.0/24	254	VoIP phones (tagged, QoS priority)
Management VLAN 99	192.168.99.0/28	14	Switch & AP management interfaces

1.6 Justification (Fault Isolation & Manageability)

- Fault isolation: because every device has its own dedicated home-run cable to the switch, a single failed cable or NIC affects only that one device — the remaining 19 users are unaffected, unlike a shared-medium design.
- Centralised troubleshooting: link lights and port statistics on the managed switch let a single technician quickly identify which port/cable is faulty, without disturbing other users.
- Growth headroom: a 24-port switch leaves 4-6 spare ports beyond the initial 20 devices, and a second switch can later be daisy-chained (uplinked) if the company grows further, extending the star into an extended star without redesigning the wiring already in place.
- Quality of Service: tagging the VoIP phones into a separate voice VLAN with QoS priority protects call quality even when data traffic (large file transfers) spikes.

1.7 Testing and Validation Plan

- Use a cable tester/certifier on every Cat6 run to confirm wire-map, length (<100 m), and Gigabit-capable performance before go-live.
- Verify DHCP scope assignment and connectivity with ping/traceroute from a sample device on each VLAN to the gateway and to the internet.
- Simulate a single cable failure (unplug one workstation) and confirm on the switch's port/LED status and via ping that only that device is affected, validating fault isolation.
- Load-test the VoIP VLAN during a simulated data transfer to confirm QoS priority keeps call quality (jitter/latency) within acceptable limits.

Solution 2 — Partial Mesh Topology for a Bank

2.1 Problem Analysis

Requirements: near-zero downtime for financial transaction processing, regulatory compliance for uptime and data integrity, protection against both link and device failure, low-latency connectivity between data-center devices, and strong resistance to electromagnetic interference given dense equipment racks.

2.2 Proposed Topology and Network Diagram

A partial mesh is proposed at the core layer: core switches, core routers, and firewalls are interconnected directly with one another, while distribution and access layers connect upward to at least two core devices for redundancy (dual-homing).

```
          [Core-Router-1]======[Core-Router-2]
                |   \\         //    |
                |    \\       //     |
          [Core-SW-A]==========[Core-SW-B]
             /       \\        //      \
      [Firewall-1]  \\       //  [Firewall-2]
                      \\     //
                [Dist-SW-Floor1] [Dist-SW-Floor2]
                  (dual-uplinked to Core-SW-A & B)
                         |
                  [Access Switches -> Tellers, ATMs, Servers]

  Legend:  ====  = full-mesh fiber link between core devices
           dual uplinks = redundant path from every lower layer
```

2.3 Device and Media Specification

Item	Model / Type	Qty	Purpose
Core routers	Cisco ASR / Juniper MX series (dual)	2	BGP-based redundant internet/WAN edge
Core switches	Enterprise L3 chassis switches	2	Core mesh; inter-VLAN & OSPF routing
Firewalls	Next-gen firewall appliances (HA pair)	2	Security, redundant via failover cluster
Distribution switches	L3 switches, dual-uplinked	per floor	Aggregation from access layer
Fiber - inter-building	Single-mode fiber (SMF)	as needed	Long-distance, low-loss core links
Fiber - intra-data-center	Multi-mode OM4	as needed	High-bandwidth short links, EMI-immune
Power/backup	Redundant PSU, UPS, RAID storage	per device	Continuity during power/disk failure

2.4 Redundancy and Routing Protocols

Protocol	Layer	Purpose
OSPF	Core/Distribution routing	Sub-second rerouting on internal link failure
BGP	WAN edge (dual ISPs)	Redundant, policy-based internet connectivity
HSRP / VRRP	Core gateway redundancy	Virtual default gateway shared by Core-SW-A/B
STP / RSTP	Loop prevention	Prevents broadcast storms on redundant mesh links

2.5 Justification

- No single point of failure: every core device has at least two independent paths to every other core device, so one failed link or node cannot isolate any part of the network.
- Millisecond failover: OSPF/BGP continuously recompute shortest paths; when a monitored link goes down, traffic reroutes automatically without manual intervention, meeting the bank's near-zero-downtime requirement.
- Fiber over copper: single-mode/multi-mode fiber eliminates electromagnetic interference from banking hall equipment (UPS units, HVAC) and supports the very high bandwidth financial transaction systems require.
- Regulatory alignment: redundant power, RAID storage, and dual-path networking collectively support the uptime and data-integrity guarantees that banking regulators (e.g., RBI/PCI-DSS style frameworks) commonly require.

2.6 Testing and Validation Plan

- Perform planned link-failure drills (physically disconnecting one core fiber link) and measure failover time using continuous ping/traceroute — target sub-second to low-second convergence.
- Validate BGP failover by simulating an ISP outage and confirming automatic route withdrawal/re-advertisement to the secondary ISP.
- Run a full disaster-recovery test including firewall HA failover and RAID rebuild simulation to confirm data integrity is preserved.
- Conduct periodic penetration testing and latency benchmarking between branch/distribution switches and the core to ensure transaction-processing SLAs are met.

Solution 3 — Bus Topology for a Budget Classroom Lab

3.1 Problem Analysis

Requirements: minimum possible hardware cost, a small number of computers (typically 10-20) in one room, supervised light usage (web browsing, file sharing, no mission-critical services), and no need for a dedicated server.

3.2 Proposed Topology and Network Diagram

Two implementation options are given: the classic coaxial bus, and a modern low-cost variant using a single unmanaged switch (recommended for easier sourcing of parts today).

```
Classic coaxial bus:

[T50]---T---T---T---T---T---T---T---T---T---T---[T50]
        |   |   |   |   |   |   |   |   |   |
       PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10

  T50 = 50-ohm terminator   T = T-connector (BNC)

Modern low-cost variant:

           [8/16-port unmanaged switch]
       /   |    |    |    |    |    |    \
     PC1  PC2  PC3  PC4  PC5  PC6  PC7  PC8 ...
       (Cat5e UTP patch cable to each PC)
```

3.3 Bill of Materials (approx., 16 seats)

Item	Type	Qty	Approx. Notes
Backbone cable	RG-58 coax (or single Cat5e run)	1 run	Length of room
T-connectors + BNC	Coaxial T-piece	16	One per PC (classic option)
Terminators	50-ohm end caps	2	One at each end of the bus
Unmanaged switch (modern option)	8- or 16-port 10/100/1000	1	Replaces terminators + hub role
Patch cables	Cat5e UTP	16	PC to switch (modern option)
Teacher PC	Shares files/printer via OS file-sharing	1	Acts as informal server

3.4 Justification

- Lowest cost: eliminating a managed switch or server removes the two most expensive components of a small network, keeping the design within a tight classroom budget.
- Adequate for the use case: light, supervised, non-critical traffic (browsing, shared files) tolerates the occasional collision and does not need enterprise-grade reliability.
- Simple to deploy and understand: the flat, single-segment design is also pedagogically useful, letting students literally see how a shared-medium bus network works.

3.5 Limitations and Mitigations

- Single point of failure (whole bus down on one break) — mitigated by keeping spare coax/patch cable and terminators on hand for quick replacement, and by preferring the unmanaged-switch variant, which isolates a single bad cable to one PC instead of the whole segment.

- Collision-prone under heavy load — acceptable given the light, supervised usage pattern described; if usage grows, the room can be upgraded to a star topology later using the same PCs.

3.6 Testing and Validation Plan

- Test backbone continuity and the 50-ohm terminator resistance with a multimeter/cable tester before connecting any PCs.
- Verify each PC can reach the teacher's shared folder and the internet gateway using ping and a simple file-copy test.
- Deliberately disconnect one T-connector/patch cable to confirm the expected failure behaviour (whole bus down for coax, single PC down for the switch variant), documenting the result for student learning.

Solution 4 — Hybrid Topology for a University Campus

4.1 Problem Analysis

Requirements: thousands of users across many buildings (lecture halls, labs, libraries, dorms, admin), high redundancy needs at the backbone, manageable wiring within each building, cost-efficient labs, wireless coverage campus-wide, and painless future expansion as new buildings are added.

4.2 Proposed Topology and Network Diagram

A three-layer hierarchical hybrid design is proposed: a fiber ring/partial-mesh backbone (core), star-wired distribution to each building, and star/tree access wiring within labs and offices.

```
                  CAMPUS CORE (fiber ring / partial mesh)
      [Core-R1]====[Core-R2]====[Core-R3]====[Core-R4]
          \            |            |            /
           \___________|____________|___________/
                  (redundant 10/40 Gbps SMF links)
                       |
  -----------------------------------------------------
  |                    |                   |          |
[Dist-SW Library]  [Dist-SW Lecture Hall] [Dist-SW Dorm-A] [Dist-SW Admin]
  |   (dual fiber uplink to core, star within building)
  |
[Access-SW Floor1] [Access-SW Floor2] ...
  |                    |
[PCs/Labs/AP via Cat6/PoE]
```

4.3 Layered Device Specification

Layer	Devices	Media	Role
Core	Core routers / multilayer switches (ring or partial mesh)	Single-mode fiber, 10/40 Gbps	Campus backbone; no single link isolates a building
Distribution	One L3 distribution switch per building, dual-uplinked	Fiber to core	Aggregates each building; boundary for building-level VLANs
Access	Floor-level access switches	Cat6 UTP (and PoE for APs)	Star wiring to end devices within a building
Wireless	PoE-powered access points	PoE over Cat6	Campus-wide Wi-Fi without separate power cabling

4.4 Justification

- Redundant backbone: the ring/partial-mesh core means a single fiber cut between two buildings reroutes automatically without isolating either building, satisfying the scale and uptime needs of a large campus.
- Manageable buildings: star wiring inside each building keeps troubleshooting simple for the local IT team, exactly as in Solution 1, but scoped to one building instead of the whole campus.

- Cost-efficient labs: bus/tree wiring inside individual labs (or simple star from a cheap access switch) keeps lab costs low while still uplinking into the robust building/core structure.
- Painless expansion: adding a new building only requires a new distribution switch and a fiber run to the core — the industry-standard reason the three-layer (core/distribution/access) model is used at almost every large enterprise and campus.
- PoE for wireless: powering access points over the same Cat6 cable that carries data removes the need for electricians to run separate power lines to ceiling-mounted APs.

4.5 Testing and Validation Plan

- Test backbone redundancy by disabling one core fiber link and confirming automatic rerouting (via routing-protocol convergence time measurement) with no building losing connectivity.
- Verify each building's dual uplinks by disabling one uplink and confirming the distribution switch fails over to the second without manual reconfiguration.
- Load-test Wi-Fi coverage and PoE budget in a sample building during peak class-change hours to confirm access points remain powered and responsive.
- Run end-to-end connectivity/performance tests (ping, traceroute, iperf-style throughput tests) from a sample of access-layer devices in different buildings to the core and to the internet edge.

Solution 5 — Multi-Department Network with VLANs and Inter-VLAN Routing

5.1 Problem Analysis

Requirements: several departments (HR, Finance, Sales, IT) that each need internal communication, controlled communication between departments, reduced broadcast traffic, strong security/segregation (e.g., Finance data should not be broadcast to Sales), internet access with firewall protection, and a design that scales as more departments or staff are added.

5.2 Proposed Topology and Network Diagram

Each department is wired in a star to its own access-layer Layer 2 switch. All access switches trunk up to a central Layer 3 switch/router that performs inter-VLAN routing, with a firewall router connecting the whole network to the internet.

```
                         [ Internet ]
                              |
                      [ Firewall Router ]
                              |
                 [ Core Layer-3 Switch/Router ]
                (Inter-VLAN routing; default GW per VLAN)
               /          |             |              \
        (trunk)        (trunk)       (trunk)        (trunk)
           |              |             |               |
      [SW-HR]       [SW-Finance]   [SW-Sales]       [SW-IT]
       |   |           |   |         |   |            |   |
    PCs/Printers   PCs/Printers  PCs/Printers     PCs/Servers
    (VLAN 10)      (VLAN 20)     (VLAN 30)        (VLAN 40)
```

5.3 VLAN and IP Addressing Plan

VLAN ID	Department	Subnet	Default Gateway
VLAN 10	HR	192.168.10.0/24	192.168.10.1
VLAN 20	Finance	192.168.20.0/24	192.168.20.1
VLAN 30	Sales	192.168.30.0/24	192.168.30.1
VLAN 40	IT	192.168.40.0/24	192.168.40.1
VLAN 99	Management	192.168.99.0/28	192.168.99.1

5.4 Representative Switch/Router Configuration (Cisco IOS style)

```
! --- Access switch (e.g., SW-HR) ---
vlan 10
 name HR
interface range fastEthernet 0/1-20
 switchport mode access
 switchport access vlan 10
interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99

! --- Core Layer-3 switch/router ---
vlan 10
 name HR
vlan 20
 name FINANCE
```

```
vlan 30
 name SALES
vlan 40
 name IT
!
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
!
ip routing                    ! enables inter-VLAN routing
```

5.5 Justification

- Security through segmentation: placing each department in its own VLAN means a broadcast or a compromised device in Sales cannot directly reach Finance or HR traffic; inter-VLAN access can be explicitly restricted with access-control lists on the core router.
- Reduced broadcast traffic: broadcasts (ARP, DHCP discover, etc.) stay within a department's own VLAN/broadcast domain instead of flooding the entire company, improving overall performance as headcount grows.
- Centralised routing and internet access: the core Layer-3 switch/router handles all inter-department traffic and hands off to the firewall router for internet access, giving one clear choke point for applying security policy.
- Scalability: adding a new department is simply a matter of creating a new VLAN, assigning a subnet, and trunking a new (or existing) access switch — no rewiring of the existing departments is required.
- Redundancy-ready: redundant uplinks/backup paths (e.g., a second core switch with HSRP/VRRP) can be layered onto this same design without changing the department-facing access layer.

5.6 Testing and Validation Plan

- Confirm VLAN isolation by attempting to ping between two hosts in different VLANs before enabling inter-VLAN routing — the ping should fail, proving broadcast-domain separation.
- Enable inter-VLAN routing and re-test connectivity between departments that are permitted to communicate (e.g., IT to all departments for support), confirming routing works as intended.
- Verify ACLs by attempting a denied cross-department connection (e.g., Sales to Finance) and confirming it is blocked while permitted traffic (e.g., HR to IT helpdesk) passes.